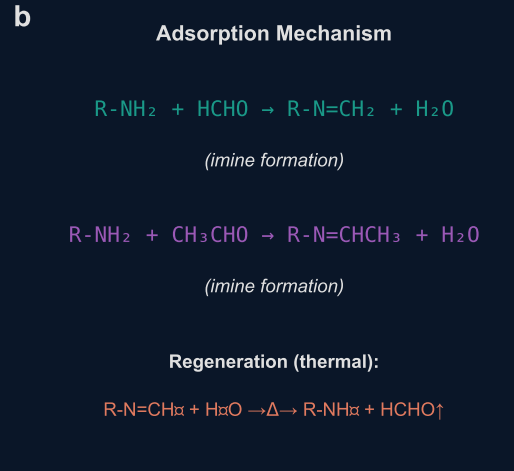
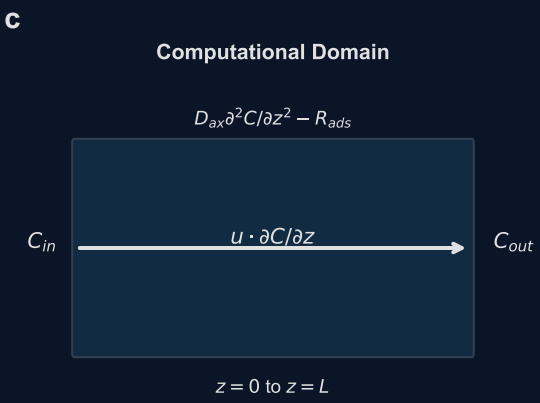
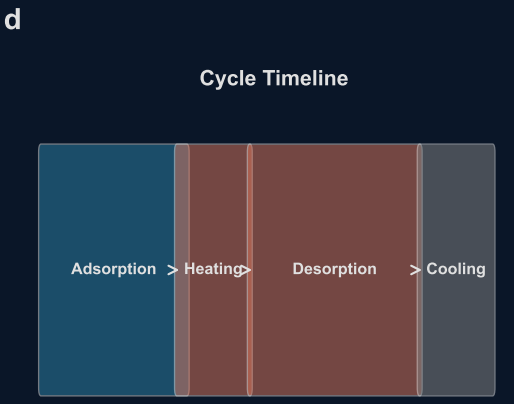
a
Cookstove Exhaust
TEG Module (Bi□Te□)
Resistive Heater
Gas flow →
Amine-Functionalized
Porous Sorbent Bed


b
Adsorption Mechanism
R-NH₂ + HCHO → R-N=CH₂ + H₂O
(imine formation)
R-NH₂ + CH₃CHO → R-N=CHCH₃ + H₂O
(imine formation)
Regeneration (thermal):
R-N=CH¤ + H¤O →Δ→ R-NH¤ + HCHO↑


c
Computational Domain
$D_{ax}\partial^2C/\partial z^2 - R_{ads}$
C_{in}
$u \cdot \partial C/\partial z$
C_{out}
$z = 0$ to $z = L$


d
Cycle Timeline
Adsorption
Heating
Desorption
Cooling